Advanced Pandas Assignment

Instructions for Students

1. Upload the dataset

Dataset:Sales_dataset into your Google Colab / Jupyter Notebook.

1. Import the Pandas library and load the dataset.
2. Write Python code to solve each question.
3. Display both code and output for every question.
4. Use appropriate Pandas functions such as filtering, grouping, reshaping, and aggregation.

Example to load the dataset:

```
import pandas as pd

df = pd.read_csv("Pandas_advanced_sales_dataset.csv")

df.head()
```

Q1 Dataset Exploration Perform the following exploratory steps: Display the first 10 rows Display the dataset structure using info() Find the number of rows and columns Identify the data types of each column

Answer:

```
import pandas as pd

# Load dataset
df = pd.read_csv("Pandas_advanced_sales_dataset.csv")

# First 10 rows
df.head(10)

# Dataset info
df.info()
```

```
# Shape (rows, columns)
df.shape

# Data types
df.dtypes
```

Explanation):

- `head(10)` shows sample data
- `info()` gives structure and missing values
- `shape` → total rows & columns
- `dtypes` → data types of each column

Q2 Sales Filtering Analysis Filter the dataset to display: Orders where Sales > 3500 Orders where Region = "East" and Sales > 2500 Explain what insights you observe.

Answer:

```
# Sales > 3500
high_sales = df[df['Sales'] > 3500]

# Region = East and Sales > 2500
east_sales = df[(df['Region'] == 'East') & (df['Sales'] > 2500)]

high_sales
east_sales
```

(Insights):

- High-value orders (>3500) indicate premium transactions
- East region with high sales may be a strong-performing market
- Helps identify target regions and high-revenue segments

Q3 Customer Purchase Analysis Identify the top 5 customers based on total sales. Hint: Use GroupBy with aggregation.

Answer:

top_customers = df.groupby('Customer Name')['Sales'].sum().sort_values(ascending=False).head(5)

top_customers

- Displays top 5 customers contributing highest revenue

Insight:

- A small group of customers often drives a large portion of sales

Q4 Regional Sales Performance Calculate the total sales generated by each region. Then identify: Which region generated the highest revenue?

Answer:

region_sales = df.groupby('Region')['Sales'].sum()

highest_region = region_sales.idxmax()

region_sales
highest_region

- Region-wise total sales
- Highest revenue region = `highest_region`

Insight:

- Helps identify strongest market

Q5 Product Performance Analysis Calculate the total quantity sold for each product. Which product appears to be the most popular?

Answer:

product_qty = df.groupby('Product')['Quantity'].sum().sort_values(ascending=False)

product_qty

- Product with highest quantity sold = Most popular product

Insight:

- High quantity ≠ high profit (important distinction)

Q6 Multi-Level Grouping Group the dataset by: Region and Category Calculate the total sales for each combination. What insights can you derive from this analysis?

Answer:

```
multi_group = df.groupby(['Region', 'Category'])['Sales'].sum()

multi_group
```

(Insights):

- Shows sales per Region + Category
- Helps identify:
  - Best-performing category per region
  - Weak segments

Q7 Pivot Table Creation Create a pivot table showing: Rows → Region Columns → Product Values → Total Sales Which region sells the most laptops?

Answer:

```
pivot_table = pd.pivot_table(df,
values='Sales',
index='Region',
columns='Product',
aggfunc='sum')

pivot_table
```

- Identify highest value under "Laptop" column → region with highest laptop sales

Insight:

- Useful for product-region comparison

Q8 Profitability Analysis Calculate the average profit for each product category. Which category is most profitable?

Answer:

avg_profit = df.groupby('Category')['Profit'].mean().sort_values(ascending=False)

avg_profit

- Category with highest average profit = Most profitable

Insight:

- High sales doesn’t always mean high profit

Q9 Discount Impact Study Analyze the relationship between discount and profit. Tasks: Calculate the average profit for each discount level Identify whether higher discounts lead to lower profit

Answer:

discount_profit = df.groupby('Discount')['Profit'].mean()

discount_profit

(Insights):

- Compare profit across discount levels
- Typically:
    - Higher discount → Lower profit

Conclusion:

- Excessive discounts reduce profitability

Q10 End-to-End Data Wrangling Task Perform the following workflow: Filter the dataset where Sales > 3000 Group data by Region and Product Calculate total sales Reshape the result into a pivot table Explain the business insights obtained from this analysis.

Answer:

```
# Step 1: Filter Sales > 3000
filtered_df = df[df['Sales'] > 3000]

# Step 2: Group by Region & Product
grouped = filtered_df.groupby(['Region', 'Product'])['Sales'].sum()

# Step 3: Pivot table
pivot_result = grouped.unstack()

pivot_result
```

(Business Insights):

- Identifies **high-performing products in each region**
- Helps in:
    - Inventory planning
    - Regional marketing strategies
- Focus on products generating high sales in specific regions